



Effects of bacterial naso-oro-pharyngeal colonisation on acute graft-versus-host disease in patients undergoing allogeneic bone marrow transplantation

KL Liu¹, R Herbrecht¹, B Lioure¹, B Jaulhac², F Maloisel¹, P Dufour¹, JP Bergerat¹ and F Oberling¹

¹Service d'Onco-Hématologie, Hôpital de Haute-pierre and ²Institut de Bactériologie, Faculté de Médecine, Strasbourg, France

Summary:

The efficacy of gastrointestinal decontamination in reducing the incidence of severe bacterial or fungal infections and of moderate to severe acute graft-versus-host disease (GVHD) has been suggested. We report here a retrospective study of 71 patients grafted consecutively in our institution with bone marrow from HLA genotypically identical siblings. Complete decontamination (plastic isolator or laminar airflow room, sterile nursing and oral antimicrobial drugs) was carried out in all patients. Sixty eight patients were evaluable. Only six patients had aerobic Gram negative rods or anaerobic bacteria in their faeces and 44 of 68 (65%) had yeasts in their faeces. Most patients had oropharyngeal and/or nasal colonisation with bacteria (Gram positive cocci: 39 patients (57%); Gram negative rods: 13 patients (19%)) or yeasts (29 patients (43%)). Thirty nine patients (57%) experienced severe grade $\geq$ II acute GVHD (grade II–IV). A significant relation was found between bacterial oropharyngeal or nasal colonisation and GVHD ($P < 0.01$) but not between gastrointestinal microflora and GVHD, whatever microorganisms were considered (bacteria, yeasts).

Keywords: naso-oro-pharyngeal colonisation; bacteria; GVHD; transplantation

During the past 20 years BMT has become a major therapeutic process for patients with aplastic anemia, severe congenital immunodeficiency diseases or hematological malignant disease. Nevertheless, complications such as GVHD and severe infections occur frequently after allogeneic BMT and may lead to death in a large number of patients. Gnotobiotic measures to prevent infections with protective isolation and suppression of the host's gastrointestinal microflora by either complete or selective gastrointestinal decontamination were demonstrated to be efficient in reducing the incidence of both infection and acute GVHD in animals^{1–5} and in humans.^{6–10} Storb *et al* found protective isolation to reduce the incidence of acute GVHD and to increase survival among patients with aplastic anemia.¹¹

Nevertheless, the protective isolation appeared less beneficial in patients grafted for leukemia.¹² Skinhøj *et al* found no influence of microbial decontamination on the incidence of GVHD¹³ and thus the correlation remains controversial.

As well as the gastrointestinal tract, the oropharyngeal cavity is a prominent site of infections, immunological reactions and toxic effects of chemotherapy after BMT such as mucositis.¹⁴ The aim of this retrospective study was to evaluate the influence of naso-oro-pharyngeal microflora on GVHD in 71 patients undergoing allogeneic BMT from HLA-identical siblings. Forty of them have been reported previously.¹⁵

Patients and methods

Patients

Seventy one patients were consecutively grafted in our institution with allogeneic marrow from HLA-identical siblings during the period October 1984 to September 1992. Patients grafted with T lymphocyte-depleted marrow from HLA-mismatched family donors or from HLA-matched but unrelated donors were excluded from this study. Three patients died early during the first 10 days after BMT and are therefore not evaluable. Death was from presumed respiratory syncytial virus pneumonia, veno-occlusive disease and massive digestive hemorrhage, respectively. Thus 68 patients were evaluable for this study: clinical characteristics are given in Table 1.

Table 1 Characteristics of 68 patients

Sex (male/female)	48/20
Age (years), median and range	30 (3–50)
Diagnosis	
ALL	21
AML	19
CML	16
Non-Hodgkin's lymphoma	3
Severe aplastic anemia	2
β -thalassaemia	2
Multiple myeloma	3
Myelodysplastic syndrome	1
Constitutional pure erythrocyte anaemia	1

Bone marrow transplantation

All patients received BM grafts from HLA-genotypically identical siblings; the mixed lymphocyte culture was negative in all cases.

Different conditioning regimens were applied, using chemotherapy alone (27 patients) or a combination of chemotherapy and irradiation (41 patients; Table 2).

GVHD prophylaxis used a combination of CsA and short-course MTX in 56 patients, or CsA alone in 12 patients.

GVHD was diagnosed on the basis of clinical symptoms and biological hepatic abnormalities, and confirmed in most patients by skin, rectal and/or hepatic biopsy. Severity of acute GVHD was graded according to Thomas *et al.*¹⁶

Prevention of infection

All patients had either a subclavian tunnelled catheter or an implantable catheter and were isolated in a laminar air-flow room or plastic isolator. Decontamination was started 7–10 days before transplantation and was given during the whole isolation period. Cutaneous mucous decontamination combined a daily bath in a chlorhexidine gluconate solution for 8 days before isolation; four times a day 0.05% chlorhexidine gluconate and polyvidone iodine mouthwashes and gargles; four times a day 0.05% chlorhexidine nasal sprays; anal applications of polyvidone iodine after each defecation; and vaginal decontamination with polyvidone iodine pessary given once daily.

Three different protocols of gastrointestinal decontamination were used in succession. (1) A combination of neomycin 3 g/day, colistin 4.5 MU/day, vancomycin in oral suspension 1.5 g/day and amphotericin B in oral suspension 4 g/day from October 1984 to October 1986 (six patients). (2) A combination of pristinamycin 3 g/day, amikacin 1.5 g/day and amphotericin B in oral suspension 4 g/day from October 1986 to September 1989 (29 patients). (3) A combination of vancomycin in oral suspension 1.5 g/day, amikacin 1.5 g/day and amphotericin B in oral suspension 4 g/day from September 1989 to September 1992 (33 patients). All patients received only sterilized food throughout the isolation period.

All patients received i.v. acyclovir prophylaxis 15 mg/kg/day started on day -1 before BMT completed with prophylactic administration of anti-CMV immunoglobulin (300–400 mg/kg weekly). CMV-seronegative patients were transfused with CMV-negative blood products. The protective isolation and gastrointestinal decontamination were generally maintained until the neutrophil count

reached $1.0 \times 10^9/l$ and was continued for approximately 1 week longer if patients were treated with high-dose immunosuppressive therapy for severe GVHD.

Surveillance cultures and treatment

Surveillance cultures for bacterial and fungal flora were performed twice weekly from nose, throat, anus, vagina, urine and faeces during the isolation period. Cultures were focused on all aerobic and facultative anaerobic bacteria, yeasts and fungi. Stool cultures were performed for both aerobic and anaerobic bacteria. All the samples were cultivated at 37°C on Drigalski agar and on trypticase soya agar with 5% fresh sheep blood. For throat swabs and sputum samples, culture was performed under 10% CO₂ on selective blood agar added with gentamicin (5 µg/ml) and on chocolate agar added with bacitracin (20 IU/ml). Vaginal swabs were cultivated under 10% CO₂ on chocolate agar added with vancomycin (3 µg/ml), colistin (7.5 µg/ml) and nystatin (12.5 µg/ml). Stool samples were cultivated on Butzler medium and at 42°C under 10% CO₂ and 5% O₂ on *Salmonella-Shigella* medium and under anaerobic conditions on blood agar added with hemin (10 µg/ml) and vitamin K₁ (1 µg/ml).

For stool and urine, cultures were assessed in a quantitative way. A positive threshold was established at 10³ bacteria per gram of stool. All other cultures were assessed as positive or negative without quantitative count.

Microbial colonisation was evaluated from day of transplantation until recovery from neutropenia and discontinuation of digestive tract decontamination. All microorganisms isolated by the cultures, even the usual mucosal flora, were taken into account for this analysis. Suppression of microflora was considered as successful if no pathogen could be isolated over the period of evaluation.

In the case of fever $\geq 38.5^\circ\text{C}$, blood cultures for bacteria and fungi were sampled. Parenteral antibiotics were given when temperature exceeded 38.5°C, and/or other signs of infection were suspected.

Statistical analysis

Two groups of patients were considered for statistical analysis: on the one hand, patients without acute GVHD (grade 0) or with only mild (grade I) acute GVHD; on the other hand patients with moderate to severe grade $\geq$ II acute GVHD (grade II–IV). The χ^2 test was used to study the correlation between microflora and GVHD, using the exact test for the small numbers.

Results

Microbiology of faecal microflora

From the day of transplantation until discontinuation of digestive tract decontamination, 66 patients (97%) had negative faecal cultures for aerobic or facultative anaerobic rods. One patient harbored *Pseudomonas aeruginosa* and *Escherichia coli* with a maximum count of 7×10^6 CFU per gram of stool and one other patient a maximum of

Table 2 Conditioning regimens

Protocol	No. of patients
Busulfan + CY	23
Total body irradiation (TBI) + Ara C + melphalan	23
TBI + CY + VP-16	12
TBI + VP-16 + melphalan	4
Busulfan + Ara C + melphalan	4
Total nodal irradiation + CY	2

2×10^6 *Pseudomonas aeruginosa* CFU per gram of stool. None of the patients had positive cultures for Gram positive cocci. Four patients (6%) harbored anaerobic bacteria (mean 3×10^6 bacteria CFU per gram of stool, range 1×10^6 to 6×10^6 bacteria CFU per gram of stool). Overall 63 patients (93%) had negative cultures for bacteria. In the case of yeasts, stool decontamination was less effective and 44 patients (65%) remained colonised by yeasts, mainly by *Candida albicans*. Overall, only 22 patients (32%) had negative cultures whatever microorganism was considered (bacteria or fungi, Table 3).

Microbiology of the naso-oropharyngeal cavity

Thirty nine patients (57%) had positive cultures once or more for Gram positive cocci during the period of evaluation. Thirty seven patients (54%) remained colonised by staphylococci (coagulase negative staphylococci alone in 29 patients, *Staphylococcus aureus* alone in four patients, both together in four patients). Eighteen patients (21%) were colonised by streptococci. Cultures were positive for Gram negative rods in 13 patients (19%). Twenty nine patients (43%) still harbored yeasts, mainly *Candida albicans*. Overall, only 17 patients (25%) had negative cultures whatever microorganism was considered (Table 3).

Infectious episodes

During the isolation period all patients experienced fever. In most cases, fever was probably related either to toxicity of the conditioning regimen or to GVHD. It is impossible to draw, *a posteriori*, any accurate conclusion about the origin of fever in those patients. An infectious origin could only be proved in 23 cases. Eleven cases of bacterial septicaemias were observed (*Staph. aureus* in six; coagulase negative staphylococci in three; *Enterococcus* in one, *E. coli* in one, and five cases of fungemia (*C. parasilopsis* in one; *C. glabrata* in three; *Trichosporon cutaneum* in one). Four cases of pneumonia occurred (*Aspergillus* in three and CMV in one). Two cases of microbiologically non-documented cellulitis and one case of cerebral toxoplasmosis were observed.

Ten of the 16 bacterial and fungal septicaemias might have originated in the naso-oropharyngeal cavity as the same microorganisms had been found only there previously. Five patients died (two *Aspergillus* pneumonias,

Table 4 Relations between gastrointestinal colonisation and acute GVHD

	Bacterial cultures		Fungal cultures	
	Positive	Negative	Positive	Negative
Patients with grade $\leq$ I GVHD	2	28	17	12
Patients with grade $\geq$ II GVHD	4	34	27	12
Difference	$P = 0.57$		$P = 0.37$	

one CMV interstitial pneumonia, one *Trichosporon* septicaemia, and one cerebral toxoplasmosis).

GVHD

Eleven patients (16%) did not develop acute GVHD. Eighteen patients (27%) experienced a grade I acute GVHD, 26 patients (38%) grade II, 11 patients (16%) grade III and two patients (3%) grade IV. Severe GVHD resulted directly in one death during the initial hospitalisation.

A statistically significant relationship was observed between a bacterial oropharyngeal and/or nasal colonisation and the incidence of moderate to severe acute GVHD ($P < 0.01$). There was no significant relationship between fungal oropharyngeal and nasal colonisation and acute GVHD. No significant relation could be established between intestinal microbial colonisation and acute GVHD whatever microorganism was considered (bacteria or fungi; Tables 4 and 5).

Discussion

As our study was initially focused on gastrointestinal flora, it rapidly became evident that most of the patients were efficiently decontaminated when considering the lower digestive tract. On the other hand, most of them still presented oropharyngeal colonisation. Therefore, we decided to analyse further the potential role of nasal and oropharyngeal colonisation in the incidence and severity of acute GVHD.

Unlike several authors,⁶⁻¹⁰ we were unable to find any correlation between incidence of acute GVHD and gastrointestinal microflora. These results are in agreement with those published by Leblond *et al.*¹⁷ and by Skinhøj *et al.*¹³ However, as most of our patients had an effective bacterial gastrointestinal decontamination, a statistical study was dif-

Table 3 Results of surveillance cultures

Microorganisms	Faeces	Oropharynx and nasal cavity
Aerobic bacteria		
Gram negative rods	2 (3)	13 (19)
Gram positive cocci	0 (0)	39 (57)
Anaerobic bacteria	4 (6)	ND
Yeasts	44 (65)	29 (43)
None	22 (32)	17 (25)

ND = not done

Values in parentheses are percentages

Table 5 Relations between naso-oropharyngeal colonisation and acute GVHD

	Bacterial cultures		Fungal cultures	
	Positive	Negative	Positive	Negative
Patients with grade $\leq$ I GVHD	12	17	14	15
Patients with grade $\geq$ II GVHD	29	10	15	24
Difference	$P < 0.01$		$P = 0.42$	

difficult to perform, but as the *P* value was 0.57, a non-significant relationship was assumed.

The study of relations between gastrointestinal colonisation and GVHD would imply the randomisation of the patients between the two groups: gastrointestinal decontamination versus no gastrointestinal decontamination. If we take into account the unanimously accepted positive effects of gastrointestinal decontamination as far as reduction of gastrointestinal infections is concerned, such a randomisation could not be ethically considered.^{10,11} A different approach could be used by comparing total and selective gastrointestinal decontamination. However, such a study would inform us only about the role of anaerobic bacteria in triggering acute GVHD. The role of anaerobic bacteria remains controversial.^{5,10} One of the suggested roles of gastrointestinal flora in the occurrence or worsening of GVHD is the release of endotoxin and bacterial antigens.^{3,18,19} According to this hypothesis, the gut represents the main reservoir of bacteria. However, there is no reason to neglect another mucosal colonisation of the organism. In cases of effective gastrointestinal decontamination, it is usual that bacteria are normally no longer found in faeces whereas the naso-oropharyngeal cavity remains more or less intensively colonised by bacteria or fungi, as in our study. Our study outlines the possible role of oropharyngeal and nasal colonisation in triggering acute GVHD.

In spite of the repeated mouthwashes and nasal pulverisations we could obtain negative culture from the oronasal cavity for only 25% of the patients. Skinhøj *et al* obtained negative culture in 35% of their patients with once to twice daily chlorhexidine washings and an intestinal decontamination based on cephaloridine or cephamandole.¹³ They mentioned no streptococci isolated in the naso-oral cavity except four cases of *Streptococcus faecalis* colonisation which may be favoured by the use of cephalosporins in the decontamination regimen.

The low rate of streptococci isolated in our patients in the oropharynx may be partially explained by the administration of vancomycin in an oral suspension which may participate in the decontamination of the upper digestive tract. Nevertheless, this appeared not to have any efficacy on staphylococci. Ten of the 16 bacterial and fungal septicaemias in this study originated apparently from remaining naso-oral microbial flora, suggesting that severe mucosal damage induced by the conditioning regimen is a major portal of entry for infections as was also suggested in other studies.^{13,14} Thus, the possibilities of local disinfection are limited and microbial proliferation, particularly of Gram positive cocci and fungi, is promoted. The relations between oropharyngeal colonisation and onset of moderate to severe acute GVHD appears clearly in our study: it could probably be demonstrated in our study because the intestine, the main reservoir of bacteria in BMT patients, has efficiently been decontaminated. Thus the role of the remaining bacterial naso-oral colonisation is yet to be revealed. In our study, most patients remained colonised by yeasts, but no relations could be shown between the incidence of acute GVHD and fungal colonisation.

These results should lead to evaluation of new methods of oropharyngeal and nasal disinfection. The local application of vancomycin as mouth washes seems to be

efficient in reducing the incidence of infection with Gram positive cocci.²⁰ The use of mupirocin which is particularly effective as far as staphylococci are concerned may be another approach.²¹

In summary, our study was initially focused only on gastrointestinal microflora, but no significant correlation could be demonstrated between acute GVHD and gastrointestinal microflora as a whole or according to the type of considered microorganism. These negative results, as opposed to the experimental data, may be in part explained by the efficiency of gastrointestinal decontamination in our study. However, we noted that oropharyngeal colonisation remains in most patients. These findings, first mentioned in the literature, allowed us to stress a correlation between the incidence of acute GVHD and the presence of bacterial oropharyngeal and/or nasal colonisation. Nevertheless our study was a retrospective study. Considering the numerous factors which may also influence the occurrence of GVHD, a prospective study should be conducted in an homogeneous group of patients grafted with the same conditioning regimen and GVHD prophylaxis. Our results may also encourage the design of new protocols for decontamination of the naso-oral cavity, as the present ones still remain somewhat disappointing.

References

- 1 Guiot HFL *et al*. Selective antimicrobial modulation of the intestinal flora of patients with acute non-lymphocytic leukemia: a double-blind, placebo-controlled study. *J Infect Dis* 1983; **147**: 615–623.
- 2 Jones JM, Wilson R, Bealmeier P. Mortality and gross pathology of secondary disease in germfree mouse radiation chimeras. *Radiat Res* 1971; **45**: 577–580.
- 3 Van Bekkum DW *et al*. Mitigation of secondary disease allogeneic mouse radiation chimeras by modification of the intestinal microflora. *J Natl Cancer Inst* 1974; **52**: 401–404.
- 4 Van Bekkum DW, Wagemaker G, Vriesendorp HM. Mechanisms and avoidance of graft versus host disease. *Transplant Proc* 1979; **11**: 189–195.
- 5 Vriesendorp HM, Heidt PJ, Zurcher C. Gastrointestinal decontamination of dogs treated with total body irradiation and bone marrow transplantation. *Exp Hematol* 1981; **9**: 904–916.
- 6 Buckner CD *et al* and the Seattle Marrow Transplant Team. Protective environment for marrow transplant recipients: a prospective study. *Ann Intern Med* 1978; **89**: 893–901.
- 7 Mahmoud HK *et al*. Laminar airflow versus barrier nursing in marrow transplant recipients. *Blut* 1984; **49**: 375–381.
- 8 Navari RM *et al*. Prophylaxis of infection in patients with aplastic anemia receiving allogeneic marrow transplants. *Am J Med* 1984; **76**: 564–572.
- 9 Schmeiser T *et al*. Application of antimicrobial prophylactic treatment to the prevention of infection and graft-versus-host disease in allogeneic bone marrow transplantation. *Exp Hematol* 1984; **12** (Suppl.15): 105–106.
- 10 Vossen JM, Heidt PJ. Gnotobiotic measures for the prevention of acute graft-vs-host disease. In: Burakoff SJ, Deeg HJ (eds). *Graft-vs-host disease: Immunology, Pathophysiology and Treatment* New York: Marcel Dekker, 1990: 403–413.
- 11 Storb R *et al*. Graft-versus-host disease and survival in patients with aplastic anemia treated by marrow grafts from HLA-identical siblings. Beneficial effect of a protective environment. *New Engl J Med* 1983; **308**: 302–307.

- 12 Storb R *et al.* Methotrexate and cyclosporine compared with cyclosporine alone for prophylaxis of acute graft-versus-host disease after marrow transplantation for leukemia. *New Engl J Med* 1986; **314**: 729-735.
- 13 Skinhøj P *et al.* and the Copenhagen Bone Marrow Transplant Group. Strict protective isolation in allogeneic bone marrow transplantation; effect on infectious complications, fever and graft-versus-host disease. *Scand J Infect Dis* 1987; **19**: 91-96.
- 14 Heimdahl A *et al.* The oral cavity as a port of entry for early infections in patients treated with bone marrow transplantation. *Oral Surg Oral Med Oral Pathol* 1989; **68**: 711-716.
- 15 Liu KL *et al.* Effets de la flore bactérienne nasale et oropharyngée sur la maladie du greffon contre l'hôte après allogreffe de moelle osseuse. *Méd Mal Infect* 1993; **23**: 658-662.
- 16 Thomas ED *et al.* Bone marrow transplantation. *New Engl J Med* 1975; **295**: 832-843.
- 17 Leblond V *et al.* Interest of laminar airflow for prevention of GVHD and infections in BMT for leukemia and lymphoma. *Bone Marrow Transplant* 1987; **2** (Suppl. 1): 181.
- 18 Heit H *et al.* Allogeneic bone marrow transplantation in conventional mice: effect of antibiotic therapy on long-term survival of allogeneic chimeras. *Blut* 1977; **35**: 143-153.
- 19 Heidt PJ *et al.* Endotoxin concentrations in the faeces of completely and selectively decontaminated children treated with bone marrow transplantation for leukaemia or severe aplastic anemia. In: *Gnotobiology and Its Applications*. Lyon: Fondation Marcel Mérieux, 1988: 178-180.
- 20 Borthen L, Heimdahl Á, Nord CE. Comparison between two non-absorbable antibiotic regimens for decontamination of the oropharynx. *Infection* 1984; **12**: 349-354.
- 21 Casewelle MW, Hill RLR. Elimination of nasal carriage of *Staphylococcus aureus* with mupirocin (pseudomonic acid): a controlled trial. *J Antimicrob Chemother* 1986; **17**: 365-372.